

Project Design Phase

Solution Architecture

Field	Details
Team Lead	Govindu Kavya Sri
Team Members	Eswari Gayatri Budumuru, Kiladi Kalavathi, Adaka Mounika, Panditi Jyostna
Domain	Retail / Cosmetics Analytics
Tools Used	Python, Microsoft Excel, Tableau Public
Dataset	Cosmetics product dataset — 1,472 records, 11 attributes
Published Dashboard	public.tableau.com/app/profile/govindu.kavya.sri/viz/cosmeticproject_17872016666220/Dashboard1

Project Context

The cosmetics industry is one of the fastest-growing and highly competitive sectors, driven by changing consumer preferences, diverse skin types, brand innovation, and evolving pricing strategies. With increasing demand for personalized beauty products, companies rely on data-driven insights to understand market trends, consumer behavior, and product performance.

This project, “Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau,” focuses on analyzing a cosmetics dataset to uncover meaningful patterns and actionable insights. The dataset includes key attributes such as brand, product category, price, rank, and skin type suitability (Dry, Oily, Combination, Normal, and Sensitive).

This project is designed to help businesses and analysts understand cosmetics market trends and consumer preferences. By analyzing brand performance, pricing patterns, product distribution, and skin-type coverage, the dashboard enables stakeholders to identify high-performing products, target specific customer segments, and optimize marketing and product strategies.

Scenario 1: Monitoring Consumer Preferences

In a real-time scenario, imagine receiving an alert indicating a concerning trend in consumer preferences, such as a significant decline in interest in certain cosmetic products or ingredients. Using the Cosmetic Insights data, we can promptly assess the extent and potential impact of this trend, identify contributing factors, and deploy immediate interventions to adapt product offerings and marketing strategies. Whether it's through targeted promotional campaigns, adjustments in product formulations, or personalized

recommendations, real-time analysis enables agile decision-making and proactive measures to meet evolving consumer needs.

Scenario 2: Addressing Product Concerns

In the event of identifying widespread product concerns, such as negative reviews or safety issues associated with specific cosmetic items, real-time access to Cosmetic Insights data enables swift response and management. Cosmetic companies and regulatory bodies can utilize the dataset to gather crucial information about the concerns, including their prevalence, potential impacts on consumer trust, and affected product demographics. By leveraging real-time analytics, they can implement quality control measures, recall products if necessary, and communicate transparently with consumers to address their concerns and maintain brand integrity.

Scenario 3: Predictive Analysis and Product Innovation

Leveraging predictive analytics capabilities, Cosmetic Insights empowers companies to anticipate and respond to emerging trends and consumer preferences in the beauty industry. By analyzing historical data and identifying predictive indicators, companies can proactively innovate new products, adjust existing formulations, and tailor marketing strategies to meet evolving consumer demands. Real-time monitoring of market trends, consumer feedback, and competitor activities enables timely interventions, product innovation, and strategic decision-making to stay ahead in a competitive market landscape.

Solution Architecture

The end-to-end architecture for this project follows a linear BI pipeline from raw data to published dashboard:

- Layer 1 — Data Source: raw cosmetics.csv (1,472 records, 11 columns — Label, Brand, Name, Price, Rank, Ingredients, and five skin-type flags)
- Layer 2 — Data Preprocessing: Python (Pandas) title-cases brand names, replaces scrape-artifact ingredient placeholders with missing values, converts a Rank of 0 to missing (unrated), and renames the skin-type columns; outputs cosmetics_cleaned.xlsx
- Layer 3 — Tableau Data Connection: Excel workbook connected to Tableau; field types verified (Price and Rank as numbers, Label/Brand/Name/Ingredients as text)
- Layer 4 — Data Reshaping: the five skin-type columns pivoted into two long-format fields, Skin_Type and Suitable, expanding the working table from 1,472 to 7,360 rows
- Layer 5 — Visualization Layer: four worksheet charts — Avg Price by Category, Price vs Rating, Top Brands by Category, Skin Type Coverage
- Layer 6 — Dashboard Layer: a single 1366×768 dashboard combining all four worksheets, with the Avg Price by Category chart set as a filter action and a searchable Brand filter control added
- Layer 7 — Publishing & Delivery: workbook titled “Cosmetic Insights: Trends & Consumer Behavior” and published to Tableau Public via File → Save to Tableau Public, generating a shareable public URL